Isabella P. Fallon PhD – Massachusetts Institute of Technology, Material Science and Engineering
• E-mail: ipf@mit.edu • Publications: google scholar
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RESEARCH INTERESTS

My research advances our understanding of the neural mechanisms underlying motor control and decision making. Combining expertise in neural circuit manipulation, activity monitoring, and continuous behavioral tracking, I developed a model of how basal ganglia circuits regulate action selection and kinematics. Currently, I am translating these insights into biologically inspired technologies that restore motor function lost to neurodegenerative disease.

EDUCATION

Massachusetts Institute of Technology, Boston, Massachusetts
Postdoctoral Associate, Material Science and Engineering
- Advisor: Polina Anikeeva

Duke University School of Medicine, Durham, North Carolina
Ph.D., Neurobiology
- Dissertation Title: Striatal Pathways for Action Counting and Steering
- Advisor: Henry Yin
- Committee: Nicole Calakos, Michael Tadross, and Rebecca Yang

University of Colorado Boulder, Boulder, Colorado
B.A., Psychology and Neuroscience
- Dissertation Title: Evaluating Stressor Controllability Effects in Female Rats
- Advisor: Steven Maier
- Committee: Kathryn Plath, Heidi Day, Michael Baratta

RESEARCH EXPERIENCE

Massachusetts Institute of Technology, Anikeeva Laboratory
Advisor: Polina Anikeeva
Postdoctoral Researcher 5/2024 – 11/2025
- Establishing a prodromal model of Parkinson's disease
- Advancing an autonomic interface for recording enteric activity
- Developing analysis pipelines for autonomic recordings
- Mentoring an undergraduate student on neuroscience, material science and experimental design

Duke University School of Medicine, Yin Laboratory
Advisor: Henry Yin
Postdoctoral Researcher 5/2024 – 11/2024
- Developed a closed-loop sensor-actuator system for reach-to-target experiments in pre-clinical models
- Implemented deep learning-based pose estimation algorithms for analyzing dexterous movements
- Measured and manipulated distinct cell populations to determine their role in dexterous movements
- Mentored PhD students on research methodologies, experimental design, and career development

Duke University School of Medicine, Yin Laboratory
Advisor: Henry Yin
Graduate Student 2019 – 2024
My research focused on the mechanisms by which the nervous system generates action. I focused on the direct and indirect pathways of the Basal Ganglia and how they contribute to action selection, kinematics and coordinating sequences of actions. My research identified a novel and simple 'push-pull' model for striatal circuits that predict action selection and movement trajectories.
- Designed a behavior task for preclinical models that improved measures of action selection and kinematics

- Measured neural activity using *in-vivo* calcium imaging and discovered distinct subpopulations
- Manipulated neural activity with optogenetics and revealed opposing contributions to behavior
- Developed a 'push-pull' neural circuit model for movement trajectories and action count

University of Colorado Boulder & UC Denver, Maier & Greenwood Laboratory
Advisor: Steven Maier
Professional Research Associate Spring 2017 – Fall 2019

- Investigated the prophylactic effects of exercise on traumatic stress

University of Colorado Boulder, Maier Laboratory
Advisor: Steven Maier
Undergraduate Research Assistant Fall 2014 – Fall 2017

- Implemented the first experiments with female preclinical rodent models in a stress lab and discovered sex differences in stress outcomes
- Leveraged pharmacology and immunofluorescence to understand neural contributions and discovered differences in dopamine signaling

Anschutz Medical Campus, Todorovic Laboratory
Advisor: Slobodan Todorovic
Undergraduate Research Assistant Summer 2016

- Explored the role of CaV3.2 channels in pain transmission and explored mechanisms of endogenous CaV3.2 channel modulation.

TECHNICAL SKILLS

- Laboratory techniques: behavioral testing & apparatus engineering (operant conditioning, pavlovian conditioning, restraint stress, shuttle box escape, juvenile social exploration, open-field, chronic unpredictable stress, water reaching), optogenetics, chemogenetics, electrophysiology, endoscopic calcium imaging, fiber photometry, pharmacology, inducible systems for activity-dependent labeling, organ baths, confocal microscopy, viral-mediated gene delivery, anatomical tract tracing, immunofluorescence.
- Data Analysis software & programming: Med-Associates programming, Python, MATLAB, Arduino C++, NeuroExplorer, Offline Sorter, Adobe illustrator, Adobe Premiere Pro, Visual Studio, Adobe Acrobat, GraphPad, ImageJ, Plexon, Minian (Ca analysis), DeepLabCut (2D&3D pose estimation), google programs.

PEER-REVIEWED JOURNAL PUBLICATIONS

1. **Fallon IP**, Fernandez SO, Hong F, Ruan S, Hollier P, Jiang Q, Yin HH. Target-dependent and bidirectional control of dexterous reaching by direct and indirect striatal pathways. (2025). ***In preparation.***
2. **Fallon IP**, Dolzani SD, Leslie NR, Amat J, Trahan GD, Laynes RA, Watkins LR, Maier SF, Baratta, MV. Coping with stress promotes long-term resilience through distinct prefrontal circuits. (2025). ***In preparation.***
3. **Fallon IP**, Roshchina M, Hong F, Fernandez SO, Ruan S, Yin HH. Striatal pathways for action counting and steering. (2024). ***In press Nature Neuroscience.***
4. Sanner K, Kawell S, Evans JG, Elekovic V, Walz M, Joksimovic SL, Joksimovic SM, Donald RR, Tomic M, Orestes P, Feseha S, Dedek A, Ghodsi SM, **Fallon IP**, Lee J, Hwang SM, Hong SJ, Maslanka M, Covey DF, Romano C, Stamenic TT, Zorumski CF, Stowell MHB, Hildebrand M, Eisenmesser EZ, Jevtovic-Todorovic V, Todorovic SM. (2024). Facilitation of CaV3.2 channel gating in pain pathways reveals a novel mechanism of serum-induced hyperalgesia. ***Under review Neuron.***
5. Bakhurin K, Hughes, RN, Jiang Q, **Fallon IP** Yin, HH. Force tuning explains changes in phasic dopamine signaling during stimulus-reward learning. (2023). ***In press Nature Communications.***
6. Tanner MK, Mellert SM, **Fallon IP**, Baratta MV, Greenwood BN. Multiple Sex- and Circuit-Specific Mechanisms Underlie Exercise-Induced Stress Resistance. (2024). ***Exercise and Mental Health.***
7. **Fallon IP**, Hughes RN, Severino FPU, Kim N, Lawry CM, Watson GDR, Roschina M, Yin HH. The role of the parafascicular thalamic nucleus in action initiation and steering. (2023). ***Current Biology.***

8. Zhang J, Hughes RN, Kim N, **Fallon IP**, Bakhurin K, Kim J, Severino FPU, Yin HH. A one-photon endoscope for simultaneous patterned optogenetic stimulation and calcium imaging in freely behaving mice. (2023). ***Nature Biomedical Engineering.***
9. Petter EA, **Fallon IP,** Hughes RN, Watson GDR, Meck WH, Ulloa Severino FP, Yin HH. Elucidating a locus coeruleus-dentate gyrus dopamine pathway for operant reinforcement. (2023). ***Elife.***
10. Hassell JE, Baratta MV, **Fallon IP,** Siebler PH, Karns BL, Nguyen KT, Gates CA, Fonken LK, Frank MG, Maier SF, Lowry CA. Immunization with a heat-killed preparation of Mycobacterium vaccae NCTC 11659 enhances auditory-cued fear extinction in a stress-dependent manner. (2022). ***Brain Behavior and Immunity.***
11. McNulty CJ*, **Fallon IP***, Amat J, Sanchez RJ, Leslie NR, Root DH, Maier SF, Baratta MV. Elevated prefrontal dopamine interferes with the stress-buffering properties of behavioral control in female rats. (2022). ***Neuropsychopharmacology.***
12. Watson GDR, Hughes RN, Petter EA, **Fallon IP**, Kim N, Severino FPU, Yin HH. Thalamic projections to the subthalamic nucleus contribute to movement initiation and rescue of parkinsonian symptoms. (2021). ***Science Advances.***
13. Frank MG, Baratta MV, Zhang K, **Fallon IP**, Pearson MA, Liu G, Hutchinson MR, Watkins LR, Goldys EM, Maier SF. Acute stress induces the rapid and transient induction of caspase-1, gasdermin D and release of constitutive IL-1β protein in dorsal hippocampus. (2020). ***Brain Behavior and Immunity.***
14. **Fallon IP**, Tanner MK, Greenwood BN, Baratta MV. (2019). Sex differences in resilience: experiential factors and their mechanisms. ***European Journal of Neuroscience***.
15. Tanner MK, **Fallon IP**, Baratta MV, Greenwood BN. Voluntary exercise enables stress resistance in females. (2019). ***Behavioural Brain Research***, 369.
16. Baratta MV, Leslie NR, **Fallon IP**, Dolzani SD, Chun LE, Tamalunas AM, Watkins LR, Maier SF. (2018). Behavioural and neural sequelae of stressor exposure are not modulated by controllability in females. ***European Journal of Neuroscience***.

CLINICAL EXPERIENCE

Duke University Department of Neurosurgery, Shadow Neurosurgery — Spring 2025

- Observed deep brain stimulation and spinal cord stimulation surgeries

Anderson Medical Center, Medical Scribe Cardiology — Summer 2017 – Fall 2017

- Documented patient encounters in real-time using electronic health records
- Transcribed physician notes, diagnoses, and treatment plans
- Maintained accurate and HIPPAA- compliant medical records

Boulder Run Physio, Shadow Physical therapy — Summer 2017 – Spring 2018

- Observed patient evaluations, treatment plans, and rehabilitation exercises
- Shadowed therapists during patient care, including manual therapy, mobility exercises, and gait training
- Gained exposure to a variety of musculoskeletal, neurological, and post-surgical cases

PEER-REVIEWED CONFERENCE ARTICLES & ABSTRACTS

1. Yin HH, Bakhurin K, **Fallon IP**, O.Jiang, M.Hossain, B.Gutkin. (Fall 2022). Deconstructing the Reward Prediction Error Hypothesis of Dopamine Function. *Society for Neuroscience,* Chicago, Illinois.
2. **Fallon IP**, Roshchina M, Hong F, Vignali C, Fernandez SO Fernandez SO, Yin HH. (Fall 2022). The Role of Striatal Neurons in Counting Behavior. *Society for Neuroscience,* Chicago, Illinois.
3. Hong F, **Fallon IP,** Yin HH. (Fall 2022). Striatal Direct and Indirect Pathway Neurons Play Complementary Roles in Action. *Society for Neuroscience,* Chicago, Illinois.
4. **Fallon IP**, Fernandez SO, Yin HH. (Fall 2022). The striatal indirect pathway resets the neural representation of numerosity. *The Assembly and Function of Neuronal Circuits,* Ascona, Switzerland.
5. **Fallon IP**, Fernandez SO, Yin HH. (Fall 2022). The striatal indirect pathway resets the neural representation of numerosity. *Society for Neuroscience,* San Diego, California.
6. Bakhurin K, Hughes RN, **Fallon IP,** Yin HH. (Fall 2022). Ventral tegmental dopamine neurons signal force vector dynamics during Pavlovian conditioning and performance. *Society for Neuroscience,* San Diego,

California.
7. McNulty CJ, **Fallon IP**, Amat J, Sanchez RJ, Root DH, Maier SF, Baratta MV. (Fall 2022). Behavioral control over stress recruits a distinct circuit in females. *Society for Neuroscience,* San Diego, California.
8. Bonar KK, Tanner MK, **Fallon IP**, Baratta MV, Greenwood BN. (Fall, 2019). Exercise effects on activation of dorsal raphe nucleus-projecting neurons during uncontrollable stress. *Front Range Neuroscience Group*, Fort Collins, CO.
9. **Fallon IP,** Levy ES**,** Dolzani SD, Leslie NR, Amat J, Trahan GD, Laynes RA, Watkins LR, Baratta MV, Maier SF. (Fall, 2019). Control over stress engages a corticostriatal projection for the production of long-term stress resilience. *Ascona Circuits Meeting,* Monte Verita, Switzerland.
10. **Fallon IP,** Tanner MK, Tamalunas AM, Baratta MV, Greenwood BN (Fall, 2019) Voluntary wheel running prevents the behavioral and neurochemical sequelae of uncontrollable stress in females. *Society for Neuroscience*, Chicago, Illinois.
11. Tanner MK, **Fallon IP**, Baratta MV, Greenwood BN. (Fall, 2018) Wheel running prevents the negative impact of stressor exposure in females. *Front Range Neuroscience Group***,** Fort Collins, Colorado.
12. Levy ES, **Fallon IP,** Baratta MV, Leslie NR, Watkins LR, Maier SF. (Fall, 2018). Determining prefrontal projections involved in the production of long-term stress resilience. *Molecular and Cellular Cognition Society,* San Diego, California.
13. Baratta MV, Dolzani SD, **Fallon IP**, Leslie NR, Amat J, Trahan GD, Laynes RA, Watkins LR, Maier SF. (Fall, 2018). Control over stress engages a corticostriatal projection for the production of long-term stress resilience. *Society for Neuroscience,* San Diego, California.
14. **Fallon IP**, Baratta MV, Leslie NR, Dolzani SD, Chun LE, Tamalunas AM, Watkins LR, Maier SF. (Summer, 2018). Behavioral and neural sequelae of stressor exposure are not modulated by controllability in females. *Neurobiology of Stress Workshop,* Banff, Canada.
15. **Fallon IP**, Baratta MV, Leslie NR, Dolzani SD, Chun LE, Tamalunas AM, Watkins LR, Maier SF. (Fall, 2017). Evaluating Stressor Controllability Effects in Female Rats. *Society for Neuroscience,* Washington D.C.

INVITED TALKS

1. **"Striatal mechanisms underlying counting behavior".** Seminar series, Duke University School of Medicine, Durham, North Carolina. 2023
2. **"Investigating the role of striatal indirect pathway neurons in quantity estimation".** Basal Ganglia Symposium, Duke University School of Medicine, Durham, North Carolina. 2022
3. "**Control over stress engages a corticostriatal projection for the production of long-term stress resilience**". Ascona Circuits Meeting*,* Monte Verita, Switzerland. 2019
4. "**Evaluating Stressor Controllability Effects in Female Rats**". Rocky Mountain Regional Neuroscience Group, University of Colorado Anschutz Medical Campus, Denver, Colorado. 2017

HONORS & AWARDS

Women Who Make a Difference **($1,000**)	2018
Biological Science Initiative Travel Grant **($800)**	2017
Best Abstract Award, Rock Mountain Regional Neuroscience Group	2017
Best Undergraduate Thesis in Neuroscience Award **($1,500)**	2017
Undergraduate Research Opportunities Program Research Assistantship Award **($1,200)**	2016
Undergraduate Research Opportunities Program Individual Grant **($1,000)**	2016
Undergraduate Research Opportunities Program- HHMI Research Grant **($2,000)**	2015
Biological Science Initiative Scholars Program Summer Award Research Grant **($500)**	2015

TEACHING EXPERIENCE

Duke University School of Medicine, Concepts in Neuroscience II TA to Prof. Kevin Franks Spring 2021

- Assisted students in understanding homework material and learning the lectures
- Presented students with methods for designing a research proposal

Duke University School of Medicine, Duke Athletics
Tuor to football and soccer student athletes Fall 2020 – Spring 2021

- Organized weekly meetings to review lecture material
- Helped students organize their notes and develop strategies for remembering material
- Educated students on topics including neuroanatomy, neurobiology, neuro-ethics, and psychology

SUPERVISION OF UNDERGRADUATE MENTEES

Duke University, URS, Sofia O. Fernandez — Spring 2022 – Fall 2024
Duke University, Pierce Hollier — Spring 2021 – Spring 2022
University of Colorado Boulder, UROP, Emily Levy — Spring 2018 – Fall 2019

SUPERVISION OF GRADUATE MENTEES

Duke University, Feiyang Hong — Spring 2021 – Spring 2024

RELATED WORK EXPERIENCE

University of Colorado Boulder, Special Undergraduate Enrichment Programs
Advisory Board Member and Website Manager — 2015 – 2016

- Promoted undergraduate research on the CU Boulder campus and facilitate faculty-student partnerships
- Managed the program's communications using the platform Web Express powered by Drupal

COMMUNITY SERVICE & LEADERSHIP

Duke University, Nucleate leadership team
Communications team — Fall 2020 – Summer 2021

- Coordinated communication between local biotech companies, researchers at institutions, students in biomedical PhDs, and the Fuqua School of Business

PROFESSIONAL DEVELOPMENT

Start MIT Martin Trust Center for Entrepreneurship — January 2026
Duke Innovation and Entrepreneurship Startup Sprint — May 2025
The Inaugural Raynor Cerebellum Project Postdoc Summit, Scottsdale, AZ — November 2024
Neurobiology Retreat, Wilmington, NC — October 2024
Accelerate to Industry, Durham, NC — May 2023
The responsible scientist II, Duke University, BIOTRAIN 754 — Spring 2023
Responsible conduct in research, Duke University, GS 710A — Fall 2020
Neurobiology Retreat, Wilmington, NC — Fall 2019
Responsible conduct in research, Duke University, GS 714 — Fall 2019